Normalization Assignment

Assignment: Telecom Database Normalization

Step-by-Step Questions and Answers

Scenario

A telecom company stores customer subscription data in one table.

customer_id	customer_name	phone_number	plan_name	plan_price	services	devices	tower_id	tower_location
C001	Aung Aung	09123456789	Premium	30000	Voice, SMS, Data	iPhone, Router	T01	Yangon
C002	Su Su	09987654321	Basic	10000	Voice, SMS	Samsung	T02	Mandalay
C003	Kyaw Kyaw	09777777777	Premium	30000	Voice, SMS, Data	Router	T01	Yangon

Part A: Questions and Answers

Q1. What normalization problems exist in the original telecom table?

Answer:
The table contains repeating groups and multiple values in one column, such as services and devices. It also mixes customer, plan and tower data in one table. This causes redundancy and update problems.

Q2. Convert the telecom table into 1NF.

Answer:
In 1NF, each column must contain atomic values.

customer_id	customer_name	phone_number	plan_name	plan_price	services	devices	tower_id	tower_location
C001	Aung Aung	09123456789	Premium	30000	Voice	iPhone	T01	Yangon
C001	Aung Aung	09123456789	Premium	30000	SMS	Router	T01	Yangon
C001	Aung Aung	09123456789	Premium	30000	Data	Router	T01	Yangon
C002	Su Su	09987654321	Basic	10000	Voice	Samsung	T02	Mandalay
C002	Su Su	09987654321	Basic	10000	SMS	Samsung	T02	Mandalay
C003	Kyaw Kyaw	09777777777	Premium	30000	Voice	Router	T01	Yangon
C003	Kyaw Kyaw	09777777777	Premium	30000	SMS	Router	T01	Yangon
C003	Kyaw Kyaw	09777777777	Premium	30000	Data	Router	T01	Yangon

Q3. What is the main issue after converting to 1NF?

Answer:
Customer, plan, service, device and tower data are repeated.

Q4. Convert the 1NF table into 2NF.

Answer:
In 2NF, remove partial dependency. Separate data that depends only on part of a composite key.

Customers

customer_id	customer_ name	phone_number	plan_id
C001	Aung Aung	09123456789	P01
C002	Su Su	09987654321	P02
C003	Kyaw Kyaw	09777777777	P01

Plans

plan_id	plan_name	plan_price
P01	Premium	30000
P02	Basic	10000

Customer_Services

customer_id	services
C001	Voice
C001	SMS
C001	Data
C002	Voice
C002	SMS
C003	Voice
C003	SMS
C003	Data

Customer_Devices

customer_id	devices
C001	iPhone
C001	Router
C001	Router
C002	Samsung
C002	Samsung
C003	Router
C003	Router
C003	Router

Customer_Towers

customer_id	tower_id	tower_location
C001	T01	Yangon
C001	T01	Yangon
C001	T01	Yangon
C002	T02	Mandalay
C002	T02	Mandalay
C003	T01	Yangon
C003	T01	Yangon
C003	T01	Yangon

Q5. What issue still exists after 2NF?

Answer:
customer_tower depends on tower_id, not directly on customer_id. This is a transitive dependency.

Q6. Convert the 2NF design into 3NF.
Answer:
In 3NF, remove transitive dependency.

Towers

tower_id	tower_location
T01	Yangon
T01	Yangon
T01	Yangon
T02	Mandalay
T02	Mandalay

Customer_Towers

customer_id	tower_id
C001	T01
C002	T02
C003	T01

Q7. Is the telecom database now in BCNF?

Answer:
Yes, if every determinant is a candidate key.
Example:

- customer_id → customer_name, phone_number, plan_id
- plan_id → plan_name, plan_price
- tower_id → tower_location

Q8. What 4NF issue can exist in telecom data?

Answer:
A customer can have multiple independent services and multiple independent devices. These should not be stored together in one table because they create unnecessary combinations.

Wrong design:

customer_id	services	devices
C001	Voice	iPhone
C001	Voice	Router
C001	SMS	iPhone
C001	SMS	Router
C001	Data	iPhone
C001	Data	Router

Correct 4NF design:
Customer_Services

customer_id	services
C001	Voice
C001	SMS
C001	Data

Customer_Devices

customer_id	devices
C001	iPhone
C001	Router

Q9. What is a possible 5NF issue in telecom data?

Answer:
A complex relationship may exist among customers, towers and usages.

Wrong design:

customer_id	tower_id	usage
C001	T01	Primary
C001	T01	Backup
C002	T02	Primary
C003	T01	Roaming

If customer-tower, customer-usage, and tower-usage relationships are independent, they can be separated.

Customer_Tower

customer_id	tower_id
C001	T01
C002	T02
C003	T01

Customer_Usage

customer_id	usage
C001	Primary
C001	Backup
C002	Primary
C003	Roaming

Tower_Usage

tower_id	usage
T01	Primary
T01	Backup
T02	Primary
T01	Roaming

Q10. What is the final normalized telecom database design?

Answer:
Final tables:

1. Customers
2. Plans
3. Towers
4. Customer_Services
5. Customer_Devices
6. Customer_Towers
7. Optional: Usage, Customer_Usage, Tower_Usage